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Assessment of the soil fertility index (SFI) for sustainable nutrient management in rice-based cropping systems of Barind tract of West Bengal, India

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Abstract

Assessing soil fertility is essential for efficient land management, especially for the ever-growing global population and shrinking arable land per person. This study aims to assess the soil fertility through the Soil Nutrient Index (SNI) and Soil Fertility Index (SFI), and also determine site-specific NPK requirements of rice-based cropping systems of the Barind Tract. The assessment was based on 12 key soil parameters: pH, electrical conductivity (EC), organic carbon (OC), essential macronutrients (N, P, K and S) and micronutrients (Fe, Cu, Zn, B and Mn). SFI and SNI were developed, and nutrient requirements were calculated using FertiliCalc software. The results indicated that 45.41% of the area had low fertility and 54.49% had moderate fertility levels. To achieve optimal rice yield (5200 kg/ha), the average nutrient requirement was 114 kg/ha (N), 14 kg/ha (P), and 50 kg/ha (K). The key findings of this study revealed that the overall soil fertility status of this soil was deficient and unable to support sustainable rice production and suggested the dosages of nutrients help build up the soil OC content and essential primary nutrients (NPK) for specific soils, and also recommended some endemic and high-yielding varieties with a scoped fertilization strategy for achieving the long-term soil fertility and sustainability of the rice-based cropping system of Barind tract. This research provides valuable insights for farmers, researchers, and policymakers to improve crop productivity, optimize fertilization, and ensure sustainable soil fertility through regular monitoring and adaptive management.

Keywords Soil fertility index, Sustainable nutrient management, Soil nutrient index, Barind tract, Rice-based cropping system, Dakshin dinajpur district

1 Introduction

Soil fertility is the soil's ability to supply essential nutrients for optimal plant growth, and understanding it is crucial for effective management, particularly in applying fertilizers to replenish nutrients depleted by crops ([191,12, 26, 47, 110]. The rapid increase in global population necessitates the efficient use of land to guarantee food security and

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safeguard environmental quality [60, 183]. To meet this challenge, focusing on enhancing agricultural productivity rather than expanding cultivated lands is more sustainable. Understanding and improving soil fertility is key in this effort, as healthy soil is the foundation of good crop yields [51, 146]. By focusing on making better use of existing agricultural resources, we can meet the rising demand for food without compromising environmental health [12, 49, 125]. The study investigates spatial variations in soil fertility parameters under rice-based cropping systems and how do they affect sustainable nutrient management practices compared to conventional fertilization in enhancing soil fertility and sustaining rice-based crop productivity over time in the Barind tract of Dakshin Dinajpur District. An assessment of soil fertility for the rice-based cropping system in the Barind tract was carried out, focusing on critical soil chemical properties such as organic carbon (OC), electrical conductivity (EC), and pH due to their significant impact on crop yield and quality [102, 117]. The assessment also included macronutrients like nitrogen (N), phosphorus (P), potassium (K), and sulphur (S), given their high utilization rates, alongside essential micronutrients like manganese (Mn), copper (Cu), zinc (Zn), and iron (Fe), which are vital for plant physiological functions and nutrition [164, 177]. Understanding soil nutrient reserves in relation to crop requirements forms the foundation of sustainable agricultural practices, which aim to safeguard soil health while ensuring consistent productivity [187, 190]. Several methods have been proposed for evaluating soil fertility, among which the Soil Nutrient Index (SNI) and Soil Fertility Index (SFI) are commonly used tools that offer different approaches to soil evaluation [16, 105]. While several studies have utilized soil fertility indices, there is limited application of location-specific SFI models for the Barind Tract. Besides that, most studies assess fertility using basic nutrient availability but do not integrate SFI with site-specific nutrient management (SSNM) of rice-based cropping systems along with detailed mapping [66]. Additionally, many fertility assessment studies do not utilize GIS-based mapping and geostatistical techniques to develop zonal nutrient management strategies for the Barind area. Here lies the unique identity and gap-filling of the research. The SNI focuses on assessing individual soil parameters and provides a straightforward and user-friendly method for nutrient management. On the other hand, the SFI employs a more integrative approach by combining multiple soil parameters into a single calculation, aiming to deliver a comprehensive and holistic evaluation of soil fertility [29, 38, 168]. Soil properties are dynamic, changing over time and across locations due to the combined influence of biological, physical, and chemical properties. Understanding these properties and their spatial variations is essential for crafting site-specific nutrient strategies [52] Panday et al., 2019; [113]. Such approaches help optimize nutrient utilization, enhance crop yields, and reduce environmental risks [80, 95, 132]. Excessive fertilizer application depletes soil fertility and contributes to environmental pollution. Therefore, adopting sustainable and targeted nutrient management practices is critical to addressing these challenges effectively [69, 123, 146]. Fertilizer requirement mapping is a precision agriculture approach that combines soil fertility analysis with spatial data to produce detailed maps. These maps highlight the specific fertilizer needs across different areas of a field or region, enabling more targeted and efficient application [118, 140]. The Inverse Distance Weighting (IDW) interpolation technique is a widely used method for estimating values at unsampled locations based on observed data from sampled locations [23, 75, 175, 176]. The primary objective of IDW interpolation is to deliver

precise and unbiased estimates. This method has been extensively applied in soil analysis and mapping, demonstrating its effectiveness in the field of soil science [20, 39, 96]. This method utilizes Geographic Information Systems (GIS) and related spatial technologies to promote the balanced application of fertilizers [14, 126, 134]. Doing so minimizes environmental risks such as nutrient runoff and leaching, while preventing economic losses caused by over-fertilization [2, 156–158, 184, 185].

Fertilizer requirements mapping serves as a valuable tool for optimizing resource use, improving crop yields, and sustainably maintaining soil health [123, 138, 179]. Integrating soil fertility analysis with fertilizer requirement mapping offers a comprehensive approach to nutrient management [81, 148, 150, 163]. Tailoring fertilizer applications to the specific fertility conditions of the soil enables farmers to achieve greater agricultural productivity without compromising sustainability [81, 144, 153, 166]. Furthermore, this practice helps reduce the environmental impact of farming by limiting excessive fertilizer use, thereby mitigating its adverse effects on water quality and biodiversity [123, 143, 156–158, 169]. Incorporating these practices into national and regional agricultural policies is increasingly essential for tackling global food security challenges and advancing sustainable development objectives [108, 131, 133]. Hence, the objectives of the study are.

- To know the spatial variation of soil properties using IDW interpolation techniques.
- To access the soil fertility status of the study area using SNI and SFI.
- To develop site-specific NPK requirements for sustainable rice cultivation of the Barind tract of Dakshin Dinajpur district.

The findings of the present study can help soil scientists, environmental geochemists, policymakers, researchers and farmers in predicting the behaviour of overall soil health and sustainable nutrient management of the study area. Here lies the novelty of the work in connection to the outcome of the work for the benefit of the farmer's decision-making process, which in turn is directly related to the regional crop production scenario of the Hilli block. This study's novelty lies in integrating SFI and FertiliCalc-based spatial nutrient mapping in a data-scarce region, which in turn is directly related to the regional crop production scenario of the Hilli block. The study claims a good position to deal with the minute utilization of SNI and SFI indices, along with accomplishing integrated FertiliCalc software use for explicitly finding out the NPK doses. To enhance site-specific fertility improvement in surface soil is the unique identity of the present study.

2 Materials and methodology

2.1 Description of the study area

The Barind Tract is a distinctive geographic region in the northern part of West Bengal, primarily situated in the Malda, Murshidabad, and Dinajpur districts. This region is predominantly composed of old alluvial soil, also known as Barind soil, which is generally characterized by its hardness, compactness, and reddish-brown colour, which has been represented in Fig. 1 [35]. The Barind Tract lies on the Pleistocene terrace, and its soil is derived from ancient alluvial deposits, which makes it distinct from the newer alluvium found in the adjoining river valleys [101]. The soil formation in this area is attributed to the weathering of ancient alluvial deposits, and over time, it has undergone various transformations due to climatic and geological processes [155]. Hilli Block is located in

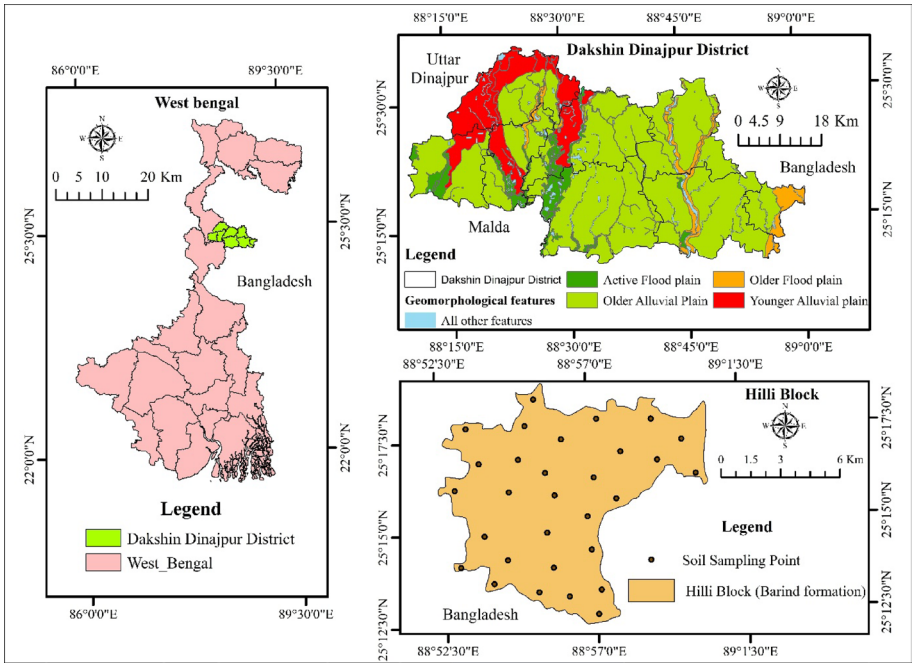


Fig. 1 Location map of the study area interpreting its geomorphological broad characters

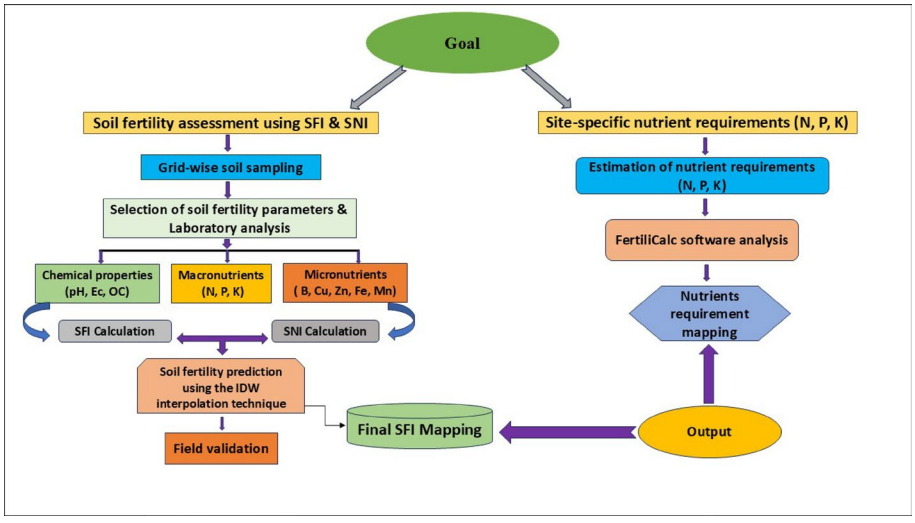


Fig. 2 Methodological flowchart for assessing SFI and site-specific nutrient requirements mapping

the southern part of the Dakshin Dinajpur district in West Bengal, India, bordering Bangladesh. It is situated between latitudes 25.3° N and 88.8° E, making it part of the northern plains of West Bengal [112] Fig. 2.

The soil in Hilli Block, typical of the Barind Tract, is primarily old alluvial soil with a clay-loam to sandy-loam texture. As per the major USDA soil classification, the Barind tract of Dakshin Dinajpur falls taxonomically in the order *Inceptisol* with a great group of *Haplaquepts* or *Eutrochrepts* with a quasi-lateritic reddish expression. The soil is compact, low in organic matter, and generally less fertile, requiring inputs such as fertilizers for successful crop cultivation [94]. Farmers in Hilli Block cultivate different varieties of rice, depending on the water availability and soil conditions. High-yielding varieties

(HYVs) are common during the monsoon season, while traditional varieties may be grown in smaller areas. The main rice-growing season in Hilli Block follows the kharif (monsoon) season, which lasts from June to October. The elevated terrain of the Barind Tract leads to challenges in accessing surface water, making the Hilli Block particularly vulnerable to drought during the dry season [139]. This soil type poses challenges for agriculture, as it is less fertile compared to the neighbouring Gangetic plains and has a lower water retention capacity. The soil of the Barind Tract is a major limiting factor for agricultural productivity in the region. Its compact, nutrient-poor nature, along with poor water retention capacity, poses significant challenges to farming (NABARD report, 2021-22). Hilli Block experiences a tropical monsoon climate, with hot summers and cool winters. The average annual temperature ranges from 15 °C in winter to 38 °C in summer, while the area receives moderate annual rainfall, primarily during the monsoon season (June to September). Rainfall averages around 1,500-2,000 mm per year, with most of it concentrated during the monsoon season [112].

2.2 Methodology and data analysis

2.2.1 Soil sampling

Grid-wise surface soil samples (0 to 20 cm) were collected from 30 locations covering a 4 × 4 sq. km grid of the study area using the fishnet method in ArcGIS software (version 0.8). Topsoil samples were collected using a hand-held auger from every grid. We excavated four samples within the same grid and mixed properly to prepare a mean sample through a 2-level quartering process and discarding the two opposite quarter sections to reach the desired weight of sample (500 g). Thus, at the field investigation stage, we collected a total of 120 samples (30 × 4), which were reduced to 30 numbers of mean samples being representative of every grid with an average distance of separation ranging from 2 to 2.5 km. Then, different representative soil sampling sites were geo-located using a handheld GPS recorder (Fig. 1). These samples were placed in a plastic tray and thoroughly mixed, followed by the quartering method. Finally, 500 g of soil was taken from each site and placed in labelled plastic bags for laboratory analysis.

2.2.2 Laboratory analysis

Before analysis, the soil sample was air-dried and crushed, and sieved through a 200-μm sieve and kept in a plastic container so that soil properties could be analyzed. The pH, EC, OC, N, P, K, S, Zn, B, Fe, Mn, and Cu were analysed following conventional methods (Table 1).

2.2.3 Soil fertility assessment

The fertility of the soil was assessed using two different methods:

2.2.3.1 Soil nutrient index (SNI) The SNI, based on the level of soil nutrient richness, was calculated according to Eq. (1) and interpreted following the standards proposed by [119] as shown in Table 2.

$$SNI = \frac{(1 * NL) + (2 * NM) + (3 * \ HM)}{NT} \tag{1}$$

Where:NL = Number of samples in Low category.

Table 1 Methods used for soil sample analysis

Soil parameters	Analysis method	References
pH	Digital pH meter in 1:2.5 (soil: water ratio)	Jacson [68]
EC (ds/m)	Digital conductivity meter in 1:2.5 (soil: water ratio)	Jacson [68]
OC %	Walkley-Black rapid titration method	Walkley and Black [174]
N(Kg/ha)	Alkaline permanganate method	Subbaiah & Asija (1956)
P (Kg/ha)	$NaHCO_3$ extractable P (pH 8.5)	Olsen et al. [114]
K (Kg/ha)	NH_4OAc extractable K	Jacson [68]
Sulphur (ppm)	Heat-soluble sulphur method	Williams & Stienbergs (1969)
Boron (ppm)	0.15% $Cacl_2$ extractable S	Berger and Truog [21]
Iron (ppm)	DTPA-extractable method	Lindsay & Norvel [89]
Manganese (ppm)		
Copper (ppm)		
Zinc (ppm)		

Table 2 Standard rating chart for soil nutrient index

Nutrient index	Interval	Category
I	< 1.67	Low
II	1.67–2.33	Medium
III	> 2.33	High

NM = Number of samples in Medium category.
NH = Number of samples in High category.
NT = Total number of samples.

2.2.3.2 Soil fertility index (SFI)

Step 1 Normalize the parameters.

Each soil fertility parameter (pH, EC, N, P, K, etc.) is often normalized or standardized to a range (commonly between 0 and 1). This ensures that different units or scales of measurement (e.g., ppm vs. percentage) don't dominate the index. The formula for Min-Max Normalization can be expressed as:

$$Normalized\ Value(\mathbf{X}\prime) = \frac{\boldsymbol{X} - \boldsymbol{X}\ min}{\boldsymbol{X}\ max - \boldsymbol{X}\ min} \tag{2}$$

Where:
X = Original value of the soil parameter.
X min = Minimum observed value of the parameter in the dataset.
X max = Maximum observed value of the parameter in the dataset.
X' = Normalized value.

Step 2 Assign weightage.

Each parameter is given a weight based on its relative importance in determining soil fertility. These weights might come from literature-based or expert surveys, which are shown in Table 3.

Step 3 Compute weighted sum.

After that, the normalized values for each parameter are multiplied by their respective weights, and then these products are summed up to get the SFI for each sample site. The

Table 3 Weightage for different soil fertility parameters

Fertility Parameters	pH	EC	OC	*N*	*P*	K	S	Zn	B	Fe	Mn	Cu
Weightage	0.15	0.05	0.2	0.15	0.14	0.14	0.05	0.03	0.02	0.03	0.02	0.02

SFI is based on the principle of the weighted sum evaluation method [151]; [85]. The formula for SFI can be expressed as:

$$SFI = \sum_{I=1}^{n} Wi * Fi \quad (3)$$

Where:

n = Number of soil parameters considered.

Wi = Weighting coefficient for each factor.

Fi = Rating or score for that factor.

Step 4 Inverse distance weighting (IDW) technique for soil fertility mapping.

For preparing the final SFI map IDW is a widely used geostatistical method for mapping soil fertility because it is easy to use and efficient to analyse. IDW assists in soil fertility mapping by using known sample points to estimate fertility levels at unsampled locations. [54, 137]. Its accuracy is determined by the significance of the weighting criterion, as well as the sampling density and interval. The IDW technique was applied to estimate localized characteristics of soil nutrients and fertility indices [18, 165]. Soil parameter maps, SFI map, and nutrient requirements maps were generated through interpolation using the Spatial Analyst extension in ArcGIS 10.8. The IDW technique was applied to estimate localized characteristics of soil nutrients and fertility indices.

2.2.3.3 Assessment of nutrient requirements Crop nutrient requirements for N, P, and K of the study area were estimated using the FertiliCalc software (http://www.uco.es/fitotecnia/fertilicalc.html). FertiliCalc assesses soil nutrient availability, analyzing essential components like N, P, and K and other relevant properties [85, 171]. FertiliCalc is a free source software developed at IAS-CSIC and the University of Cordoba to assist in NPK fertilizer doses. The system needs soil parameter values, viz. soil pH, OC (%), NPK (mg/kg), water supply condition, and soil mechanical type to be checked in condition etc. The inputs are aligned to a specific crop and its targeted yield. Thereon, by running the system, the software-based algorithm suggests an output of diagnosed doses of macro fertilizers per hectare. To assess the fertilizer requirements of the study area, 30 locations were randomly selected within the rice growing area, and the fertilizer needs for each point were established based on essential data, including potential yield and soil characteristics. The points were subsequently interpolated, and the study results were shown as fertilizer requirements maps. For demonstration purposes, rice was served as the model crop in setting parameters. The nutrient requirement estimations were aligned with a target yield of 5200 kg/ha for rice. Additionally, the model allows users to choose from three distinct fertilization strategies, each tailored to meet specific agronomic goals.

- A sufficiency approach (minimum fertilizer).
- A buildup and maintenance (reduced fertilizer) approach.
- A buildup and maintenance (maximum yield) approach.

Rice relies on key inputs such as target yield, soil nutrient status, previous crop residue management, and local fertilizer use efficiencies. The model assumes typical nutrient losses under specific water regimes, especially for flooded rice systems. Calibration is based on region-specific field data, including yield response curves, nutrient uptake, and recovery efficiency. Rice-specific adaptations include adjustments for anaerobic

soil conditions, stage-wise fertilizer splits, and varietal differences. Incorporating SSNM principles, FertiliCalc aims to enhance nutrient efficiency and optimize rice productivity sustainably.

The present study adopted the build-up and maintenance approach focused on achieving maximum yield. This calculation approach chose Urea as the N source, triple super phosphate for P, and potassium sulphate for K. This method aims to ensure nutrient supply while maintaining sustainability and minimizing excessive input use.

3 Results

3.1 Spatial distribution of soil fertility parameters

The soil properties varied across the sites, including pH, EC, OC, and essential macronutrients (N, P, K, and S) and micronutrients (Fe, Cu, Zn, B, and Mn). The soil parameters that have been observed are as follows:

The pH of the soil is referred to as the "master variable" because of the significant influence it has on a vast number of chemical reactions that involve different vital plant nutrients [43, 122]. The distribution of soil pH ranged mostly from 5.0 to 6.5 which has been shown in Fig. 3. A few samples show pH values slightly higher than 6.5, indicating somewhat alkaline conditions [109]. The spatial distribution of pH across the study area is depicted in Fig. 4a. The distribution of EC values is highly skewed, with most soils having very low conductivity between 0.1 and 0.2 ds/m (Figs. 3 and 4b). This indicates that the majority of the soil samples have low salinity, which is suitable for most crops [17, 44]. OC is crucial for soil fertility, affecting nutrient availability and soil structure [90, 98]. There is a moderate distribution in the OC percentage (Fig. 3). Soil OC levels typically range from 0.3 to 0.7%, suggesting a moderate amount of organic matter. Most of the high OC values were found in the extreme western and some areas of the southern part of the study area, as this area falls under riverbanks and floodplains that support diverse vegetation, including riparian forests, and management practices that increase

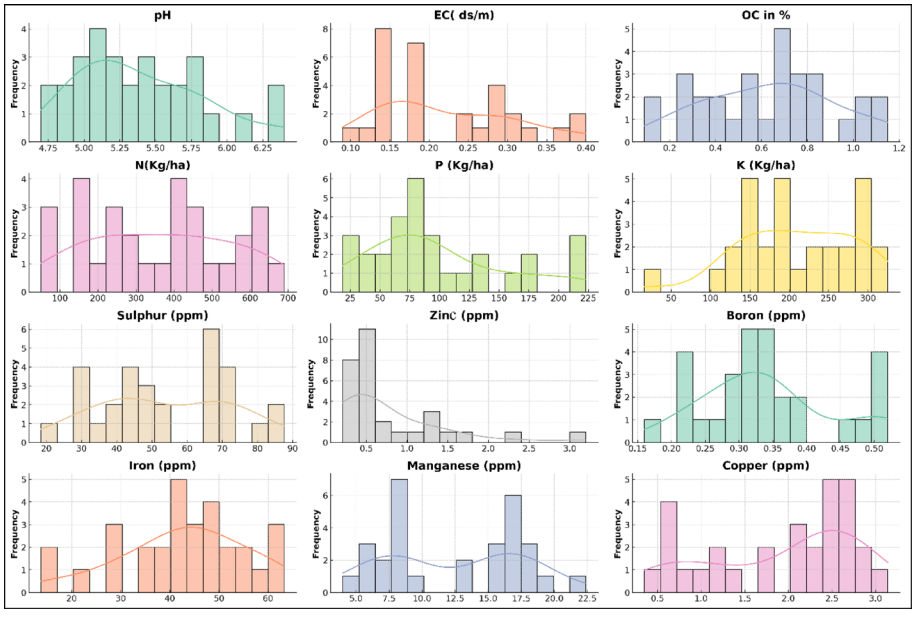


Fig. 3 Distribution of different soil properties as dominant constituents of soil fertility to identify at a glance the appearance pattern of the selected properties

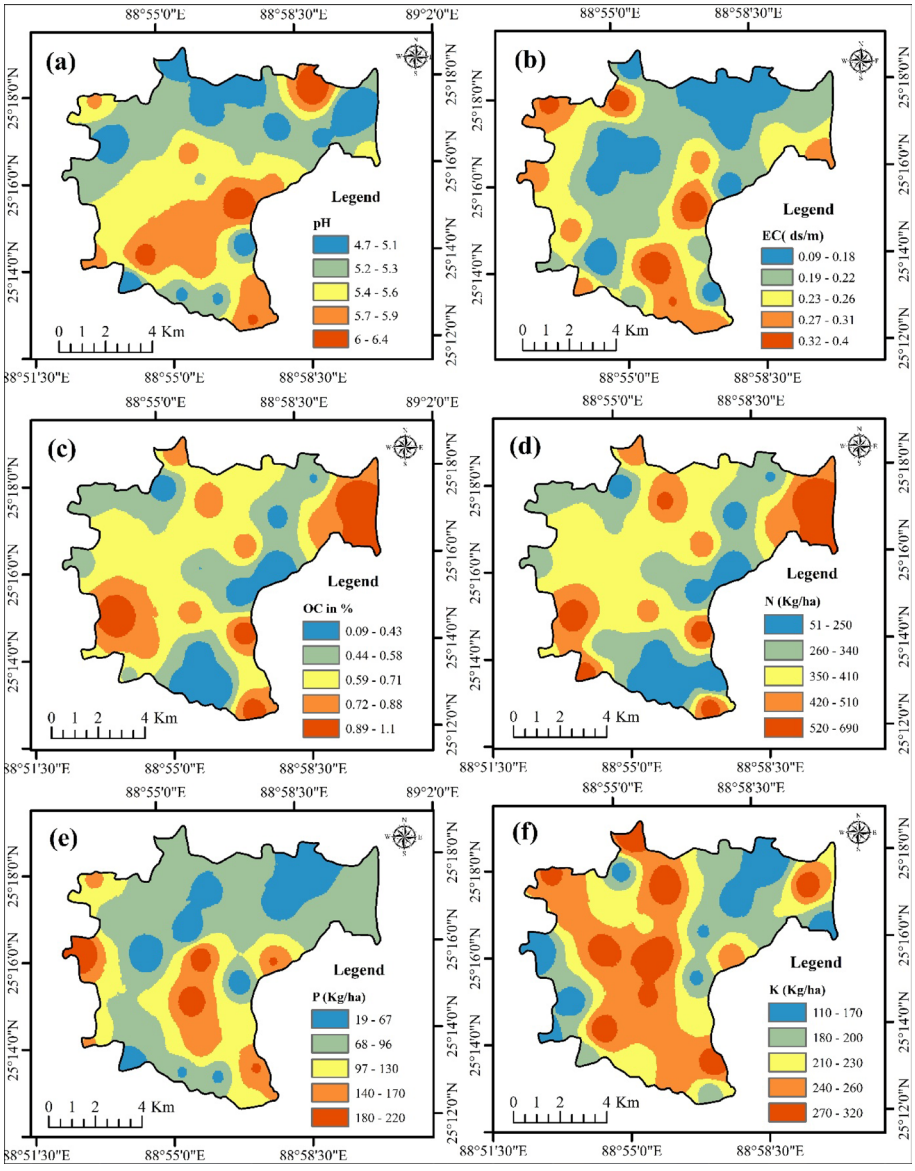


Fig. 4 Spatial distribution of different soil fertility parameters (**a**) pH, (**b**) EC, (**c**) OC, (**d**) N, (**e**) P, (**f**) K (**g**) S, (**h**) B, (**i**) Cu, (**j**) Zn (**k**) Fe and (**l**) Mn

OC concentration [42]. In contrast, the lowest concentrations are found in the northern, northeastern, and a few areas of the southwestern parts of the study area (Fig. 4c).

N is essential for plant growth, particularly for leafy vegetables [33, 59]. N distribution is somewhat spread out, with a majority of samples concentrated between 150 and 400 Kg/ha (Fig. 3). The high N content is found in the extreme western and some areas of the southern part of the study area. In contrast, the lowest concentrations are found in the northern, northeastern, and a few areas of the southwestern parts of the study area (Fig. 4d). Low N in the study region is mainly due to low to medium level of organic carbon, slightly acidic to acidic soil, leaching, low fertilizer usage, and unfavourable physical conditions like waterlogging or poor retention [28]. This suggests a variance in soil fertility, with some soils having sufficient nitrogen levels, while others may require fertilization. P is vital for root development and energy transfer in plants [116]. P levels exhibit

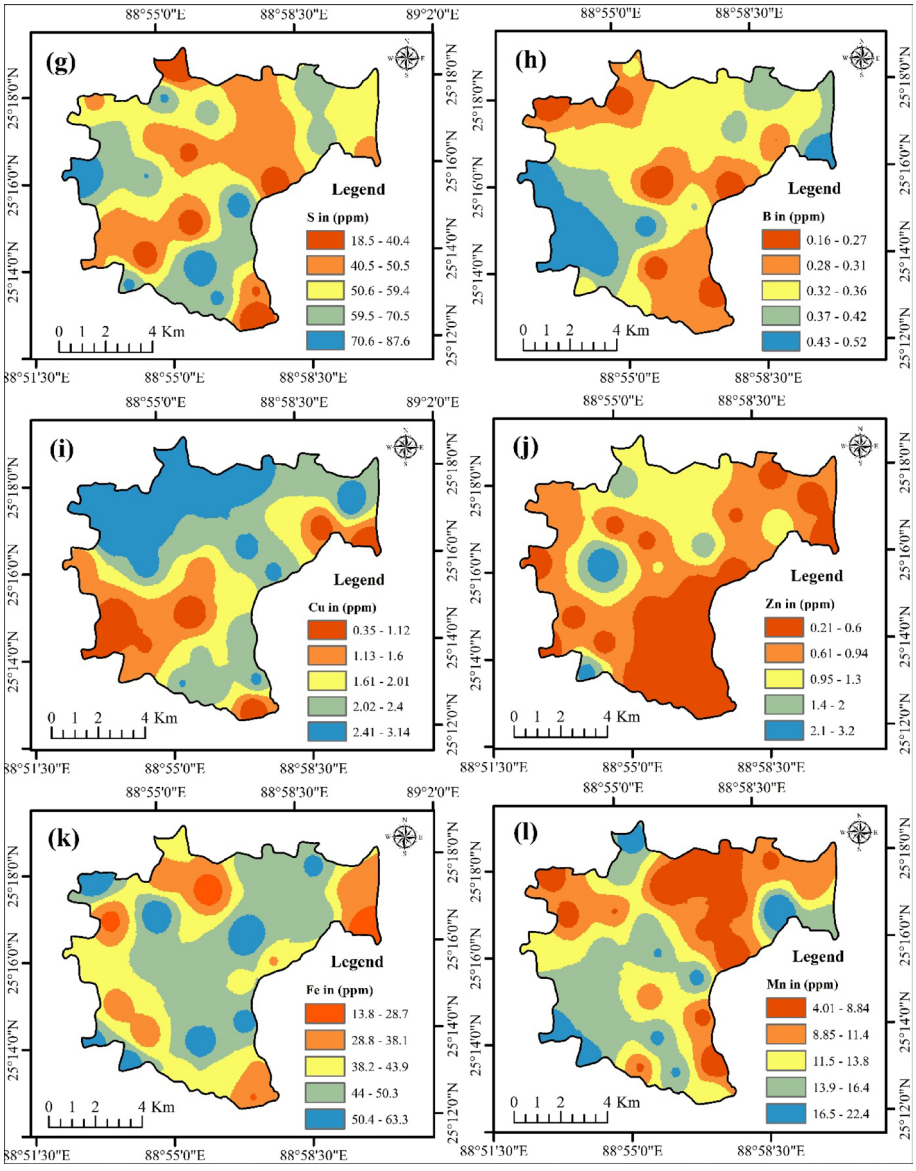


Fig. 4 (continued)

a highly spread distribution, with values ranging widely from around 20 to 50 Kg/ha (Fig. 3). This spatial variability may be linked to differential fertilization or parent material. The highest P concentrations were found in the extreme eastern and some areas of the western part of the study area. In contrast, the lowest concentrations are found in the southeastern and southern parts of the study area (Fig. 4e). Poor incorporation of farmyard manure, compost, or green manures reduces microbial activity, which may reduce the P mineralization and availability in the study area [30]. This suggests a significant variation in phosphorus availability, which might affect plant growth differently depending on the location. K plays a role in water regulation and enzyme activation in plants [62, 175, 176]. The distribution of K suggests that most samples have values between 100 and 300 Kg/ha (Fig. 3), indicating moderate to high potassium levels in the soils, which is beneficial for crop yield and plant health. The highest concentrations were found in the northern, southern, eastern, and central parts of the study area (Fig. 4f).

In contrast, the lowest concentrations are found in the northeastern and southwestern parts of the study area, as the block has intensive agriculture, often with rice-based cropping systems. Crops like paddy remove large quantities of potassium, especially when straw is removed from the field [13]. S deficiency causes growth stunting, leaf paleness or yellowing, and lower crop production and soil fertility [10, 15]. S levels show a wide range, with many samples falling between 40 and 80 ppm (Fig. 3). The highest concentrations were found in the western and southern parts of the study area. In contrast, the lowest concentrations are found in the northeastern and southwestern parts of the study area (Fig. 4g). This indicates varying S availability across the samples, which can influence plant growth and enzyme activation in crops.

B plays a fundamental role in soil fertility and plant health [136, 147]. B contributes to soil fertility by maintaining optimal pH and enhancing the microbial activity necessary for nutrient cycling [36, 72]. B shows a narrow distribution, with most samples having values between 0.35 and 0.40 ppm (Fig. 3). This is relatively consistent and suggests that boron availability is acceptable for plant growth. The highest concentrations were found in the western and southwestern parts of the study area. In contrast, the lowest concentrations are found in the extreme eastern parts of the study area (Fig. 4h). The distribution of Cu is centred around 2 ppm, with most samples showing similar levels (Fig. 4i). This suggests that Cu levels are consistent and within an acceptable range for crops [70, 147, 152]. Zn values are generally low, clustering around 0.5 ppm in most samples (Fig. 4j). Fe plays a vital role in soil fertility and crop productivity as an essential micronutrient required for plant growth and development [127, 136, 167]. Fe levels are well-distributed between 40 and 50 ppm (Fig. 3). The highest concentrations were found in the northeastern, central, and southeastern parts of the study area. In contrast, the lowest concentrations are found in some parts of the northern and extreme eastern parts of the study area (Fig. 4k). These values are within the typical range needed for plant nutrition, indicating no significant iron deficiency across the samples. Distribution of Mn shows moderate levels, with values between 7 and 17 ppm (Fig. 3). This indicates sufficient manganese levels for plant health. However, a few samples may have values at the higher end, which could indicate excess Mn (Fig. 4l).

3.2 Descriptive statistics of different soil fertility parameters

The descriptive statistics of the soil properties provide a comprehensive understanding of the variability and fertility status of the soil in the study area (Table 4). The constant interaction of natural and exogenous influences, such as the formation of soil or rock deposits, causes all of these properties to have a medium to high degree of variability. The pH values range from 4.7 to 6.4, with a mean of 5.4, indicating that the soil is generally acidic. Sd (0.45), which is relatively small, meaning most soil pH values are clustered fairly close to the mean.CV (8.29%), which means very low variability of pH or uniform across samples. skewness (0.68), indicating the data is fairly symmetric but slightly skewed toward higher pH values, and kurtosis (−0.14). This suggests values are spread more evenly around the mean without sharp peaks or heavy tails. The low mean value of EC (0.22) and Sd (0.08) indicates that most of your soil samples have quite similar salinity levels, skewness (0.58), which means most of the soil samples have lower EC values, and kurtosis (−0.64), indicating EC values are moderately spread out, but not with sharp peaks or heavy tails. and CV (37.46%) suggests a non-saline soil, suitable for rice crops;

Table 4 Descriptive statistics of different soil properties (0–20 cm) of Rice-based cropping system of Barind tract

Indicators	Min	Max	Mean	SD	CV (%)	Skewness	Kurtosis
pH	4.7	6.4	5.4	0.45	8.29	0.68	−0.14
EC (ds/m)	0.09	0.4	0.22	0.08	37.46	0.58	−0.64
OC[a] in %	0.09	1.15	0.63	0.29	45.07	0.03	−0.66
N[a] (Kg/ha)	50.4	690	363.08	189	52.06	0.02	−1.11
P[*] (Kg/ha)	18.69	223	97.32	57.82	59.42	0.83	−0.11
K[*] (Kg/ha)	16	325	208.45	72.57	34.81	−0.45	0.01
S[#] (ppm)	18.5	87.6	54.09	18.94	35.01	0.04	−1.01
Zn[#] (ppm)	0.21	3.2	0.79	0.68	86.76	2.07	4.73
B[#] (ppm)	0.16	0.52	0.34	0.1	28.3	0.49	−0.35
Fe[#] (ppm)	13.8	63.3	42.76	12.77	29.87	−0.44	−0.18
Mn[#] (ppm)	4	22.4	12.42	5.19	41.84	0.02	−1.38
Cu[#] (ppm)	0.35	3.14	1.92	0.86	44.71	−0.58	−1.17
SFI	**0.31**	**0.56**	**0.46**	**0.07**	**14.75**	**−0.25**	**−0.56**

[a]Highly limiting nutrients

[*]Moderately limiting

[#]Low limiting nutrients

at the same time, some spots may approach moderate salinity with some localised areas of slightly higher salt concentration, but no extreme salinity problems in the study area. OC content shows a mean value of 0.63%, reflecting a moderate level of organic matter, Sd (0.29), indicating OC values are very uniform across samples. Skewness (0.03) means OC is symmetrically distributed, and there are no strong outliers of very high or very low organic carbon. This suggests most soils cluster around the mean value, with balanced distribution toward higher and lower ends. Kurtosis (−0.66) indicates that OC values are more evenly spread with fewer extreme highs or lows. This means the landscape or management practices have prevented pockets of very high or very poor organic matter. CV (45.07%), which means high variation of OC content in the study area or uneven organic matter distribution, which is essential for maintaining soil structure and fertility. N content exhibits a mean (363.08 Kg/ha) and a high CV (52.06%), implying uneven nitrogen distribution, which can affect crop productivity if not managed properly. Sd (189) means nutrient distribution is heterogeneous across the field or region. Skewness (0.02) indicates that soils with low and high N content occur in nearly equal proportions around the mean. Kurtosis (−1.11) suggests that extreme nitrogen-deficient or nitrogen-rich soils are rare; most soils lie within a moderate range, but with broad variability in the study region, and also need site-specific nitrogen management rather than uniform application. P content also shows high CV (59.42%) with a mean of 97.32 Kg/ha, highlighting the need for targeted phosphorus management to avoid deficiencies or excesses. Sd (57.82) indicates the variability or spread of soil P values around the mean. This variability may come from differences in land use, fertilizer application, soil texture, or organic matter distribution within the study area. Skewness (0.83) and Kurtosis (−0.11) indicate that many soils are low to medium in P, but a few sites are highly enriched, which indicates nutrient management is not uniform in the study area. For better management, site-specific nutrient management (SSNM) is needed instead of blanket P fertilizer recommendations.

K levels are relatively high mean value (208.45 Kg/ha) and relatively higher CV (34.81%), which again suggests the importance of localized nutrient management practices. Sd (72.57) is relatively high, which means there is wide variability in K across the

study area. Skewness (–0.45) means the distribution tail is slightly towards the lower K values. Kurtosis (0.01) indicates an almost normal distribution of soil K in the study area. S levels are moderately high, with a mean of 54.09 ppm, but variations exist that warrant monitoring. Sd (18.94), indicating values across soil samples are highly variable, which means some patches in the field may have very high S while others are deficient. skewness (0.04), and Kurtosis (–1.01), indicating the distribution is almost balanced around the mean; S-rich and S-poor soils are not heavily biased one way. Thus, the distribution is generally balanced and not dominated by extreme deficiency or toxicity in the study area. The micronutrients, including Zn, B, Fe, Mn, and Cu show substantial variability, especially Zn (CV = 86.76%), indicating inconsistent availability that may require micronutrient supplementation through foliar sprays or soil amendments. The SFI has a mean value of 0.46 with moderate variation, suggesting an overall moderate fertility status in the area, though with spatial disparities. These findings underscore the need for site-specific nutrient management and soil amelioration strategies to enhance agricultural productivity and sustainability.

3.3 Correlation between different soil fertility parameters

This correlation matrix reveals complex interrelationships among soil properties, with implications for nutrient management and soil health. A strong positive correlation between OC and N ($r = 0.88$, $p = < 0.01$), which suggests that soils with higher organic carbon content tend to have higher nitrogen levels. This is expected because organic matter is a major source of nitrogen in the soil, and also a strong positive correlation between EC and S ($r = 0.53$, $p = < 0.01$). After all, Soils with higher EC tend to have higher S concentrations. EC is often linked with soluble salts, and S, being a component of such salts, tends to show higher levels in more conductive soils. However, a positive correlation between OC and B ($r = 0.44$, $p = < 0.02$) because Soils richer in OC also tend to have higher B levels, which is because of organic matter helps retain micronutrients like B in the soil and a positive correlation between N and B ($r = 0.36$, $p = < 0.05$) as N-rich soils show a positive correlation with B levels, likely due to similar soil processes affecting the retention of nutrients. The correlation between each soil property has been demonstrated in Fig. 5.

A weak to moderate negative relationship between N and pH ($r = -0.33$, $p = > 0.05$) exists because Soils with lower pH tend to have higher N levels. This may be because nitrogen-fixing microorganisms can thrive in slightly more acidic conditions, leading to an accumulation of N in such soils and a strongly negative correlation between Zn and pH ($r = -0.34$, $p = > 0.05$), as Lower pH tends to correlate with higher Zn availability. Zn becomes more soluble and available to plants in acidic soils. Similar to Zn, Cu also tends to be more available in more acidic soils, explaining the negative correlation with pH ($r = -0.32$, $p = > 0.05$). Besides this a moderate negative correlation between EC and K ($r = -0.34$, $p = > 0.05$) because higher EC seems to correlate with lower K levels. This could suggest that high salt content in the soil might inhibit the availability or uptake of K.

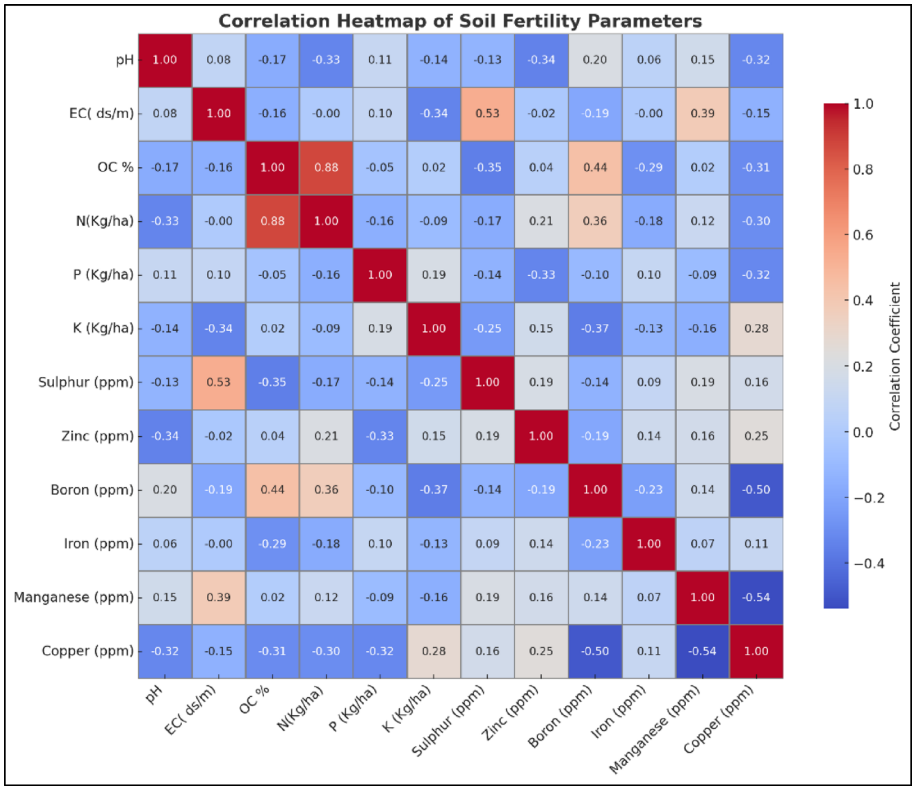


Fig. 5 Heatmap depicting the Pearson correlation matrix between the different fertility parameters of the soil in the study area

4 Discussion

4.1 Assessment of soil fertility status using environmental indices

The fertility status of the soils in the research region was analysed using the soil nutrient index and soil fertility index modelling based on various soil physicochemical parameters.

4.1.1 Soil nutrient index (SNI)

Soils in the study area were characterized by fertility categories related to pH, EC, OC, and essential macronutrients (N, P, K, and S) and micronutrients (Fe, Cu, Zn, B, and Mn). SNI values of different fertility parameters have been depicted in Fig. 6.

The results highlighted that the SNI of pH is 1.97 which indicates that the pH level of the soil is skewed toward acidic. 63% of the samples fall under the low category as the Barind soils are often classified as Ultisols or Inceptisols, tending toward lateritic conditions, which are inherently acidic, nutrient-poor, and low in base saturation, which can affect nutrient availability and microbial activity, potentially limiting crop growth [63, 99]. Paddy fields nurture some favourable microbes like nitrogen-fixing bacteria, cyanobacteria, fungi, actinomycetes, archaea, and protozoa, which are expectedly too less in the Ultisols-Inceptisols association [41, 73]. They collectively enhance soil fertility, nutrient availability, organic matter decomposition, and rice productivity, though some also contribute to methane emissions and plant diseases (Imran 65; [87]. The SNI value of EC is 3, indicating no potential risk of salt stress on crops. Hence no to very less fertilizers are required to reduce alkalinity, and producers must focus on over-leaching of salts by

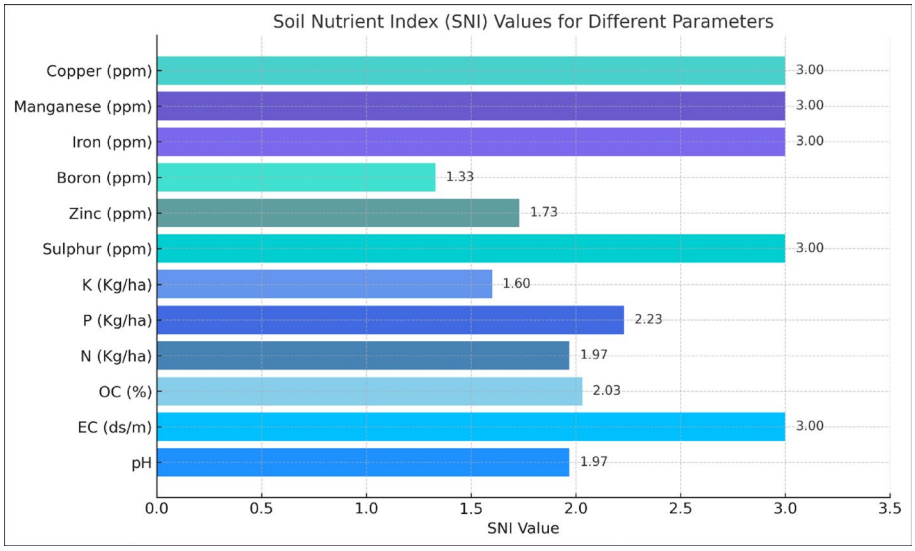


Fig. 6 Soil nutrient index (SNI) values for different soil fertility indicators

controlling the drainage wash [24]. The SNI value for OC is 2.03, suggesting moderate fertility of the soil; OC content is adequate but needs regular enrichment for sustainable fertility. Sandy and sandy loam to loam soils have less clay to protect organic matter, so C is mineralized at a faster rate in the study area [82, 170]. Higher organic carbon can improve soil structure, water retention, and nutrient supply [19, 160]. To boost the OC content, compost, farmyard manure, green manure, or crop residues to maintain and slowly increase OC levels over time [46, 91]. SNI value of N is 1.97 indicating that the soil is nitrogen-deficient in a significant portion of the samples. As a result, it reduced crop growth, as nitrogen is critical for plant development and chlorophyll production [22]. Besides that, the high temperature of the study region speeds up the breakdown of organic matter, releasing NH_4^+/NO_3^- that is then lost quickly during rains, and also high cropping intensity (2–3 crops/year) is also responsible for removing N than is replenished [64]. However, nitrogen is highly mobile and prone to leaching [31, 67]. To reduce loss of N, farmers should split the doses of urea, ammonium nitrate and combine with organic sources like compost [7]; [154]. The SNI value of P is 2.23 suggests that P availability is relatively high, with 40% of samples in the high category. Adequate phosphorus levels are important for root development, energy transfer, and flowering in crops [71, 103]. Moderate levels of P are generally adequate, but phosphorus availability declines in acidic soils because weathering, leaching, Fe/Al fixation, Ca precipitation, low organic matter, and nutrient mining reduce the amount of P and to enrich the P content in the soil, application of SSP, DAP, or bone meal, especially as a basal dose during sowing, is required for sustainable rice cultivation [4, 5]. K levels are primarily in the medium range, and the SNI value 1.6 indicates that K might be a limiting factor in crop growth. A critical deficiency is observed in this study area. Moreover, K helps in enzyme activation, water regulation, and disease resistance. Low K may result in poor grain quality, weak stems, and stress sensitivity [156–158]). Application of MOP (Muriate of Potash) or SOP (Sulphate of Potash). Wood ash is a good organic alternative [32, 45]. S levels are entirely in the high category, with an SNI of 3. This indicates that S is abundantly available in the soil. High S levels support protein synthesis and improve crop quality, particularly in oilseed crops [9, 124]. No additional supplementation is needed. Farmers should maintain

the levels through organic matter and avoid excessive application of S [25, 53]. Soil fertility rating based on SNI values of different fertility parameters has been shown in Table 5.

Zn levels are predominantly low category with an SNI value of 1.73, which can negatively impact crop yield, as it is necessary for plant enzyme function, hormone regulation, and disease resistance [103]. Use zinc sulfate, chelated zinc, or foliar sprays to meet crop demands of the barind soil [104]; [8]. B deficiency is evident in 67% of the samples, and the low SNI of 1.33 suggests that B could be a significant limiting factor in crop production. B is essential for cell wall formation and reproductive development in plants [83, 147]. Application of borax, solubor, or boronated complex fertilizers in small, monitored doses is required for sustainable rice cultivation [27, 76]. Fe is abundant in the soil with an SNI of 3, and all samples fall in the high category. Fe is vital for photosynthesis and nitrogen fixation, especially in leguminous crops [147]. Mn is present in high amounts across all samples, similar to Fe. The SNI value of 3 suggests no limitations due to Mn deficiency. Mn supports enzyme systems involved in plant metabolism and photosynthesis [31, 103]. Cu levels are also high, with an SNI of 3. High Cu availability is important for plant respiration and lignin synthesis, providing structural support to plants [88, 135, 145].

4.1.2 Soil fertility index (SFI)

Another approach to assessing the fertility status of soils is based on the calculation of the soil fertility index. The SFI is a quantitative measure that evaluates soil health based on various soil parameters [182, 186]. In this analysis, the SFI values were calculated using the weightage sum method, considering the relative importance of each soil parameter. The normalized values are shown in Table 6 and Fig. 7.

The normalized values were multiplied by their respective weights, and the results were summed to generate the overall SFI mapping for each sample site (Fig. 8). The results showed that the SFI ranged from 0.25 to 0.64 with a mean of 0.47, indicating a low to medium fertility level, which has been depicted in Table 7 and 8.

Baigram, Panjul, Basanta, Chakgopal, Purba Kalikapur, Chak Mohan, Dabra, Rupbati, Gosaipur, Purba Mustafapur, Fajullapur, Purba Mathurapur, Jamalpur, Laskarpur villages

Table 5 Soil fertility rating based on SNI values of different fertility parameters and recommended fertilizer for the study area.
Source: [11, 58]

Fertility Parameters	SNI	Soil fertility	Fertilizer Recommendations
pH	1.97	Low	Lime and Gypsum
EC (ds/m)	3	High	NA
OC %	2.03	Medium	Compost and Manure
N(Kg/ha)	1.97	Medium	Urea, Ammonium Nitrate, and Organic Fertilizers
P (Kg/ha)	2.23	Medium	Single Super Phosphate (SSP), Di-ammonium Phosphate (DAP), and Bone Meal
K (Kg/ha)	1.6	Low	Muriate of Potash (KCl), Sulphate of Potash (SOP), and Wood Ash
Sulphur (ppm)	3	High	NA
Zinc (ppm)	1.73	Medium	Zinc Sulfate, Chelated Zinc and Foliar Application
Boron (ppm)	1.33	Low	Borax or Solubor, Boronated Fertilizers
Iron (ppm)	3	High	NA
Manganese (ppm)	3	High	NA
Copper (ppm)	3	High	NA

Table 6 Normalized values of different soil fertility parameters and their weightage for SFI model

Sample sites	Chemical properties			Macronutrients				Micronutrients				
	pH	EC (ds/m)	OC %	*N* (Kg/ha)	*P* (Kg/ha)	K (Kg/ha)	Sulphur (ppm)	Zinc (ppm)	Boron (ppm)	Iron (ppm)	Manganese (ppm)	Copper (ppm)
S_1	1	0.161	0.311	0.289	0.045	0.293	0.724	0.097	0.639	0.778	0.241	0.699
S_2	0.176	0.129	0.255	0.184	0.147	0.39	0.331	0.11	0.611	0.653	0.176	0.527
S_3	0.588	0.161	0.528	0.518	0.005	0.648	0.238	0.067	0.5	0.572	0.665	0.663
S_4	0.412	0.194	0.566	0.561	0	0.891	0.757	1	0.556	0.676	0.616	0.846
S_5	1	0.968	0.189	0.175	0.074	0.452	0.908	0	0.556	0.549	0.746	0.491
S_6	0.176	0.194	0.547	0.54	0.31	0.579	0.362	0.338	0.444	0.679	0.076	0.892
S_7	0.294	0.161	0.453	0.432	0.32	0.596	0.344	0.054	0.528	0.982	0.255	0.832
S_8	0.235	0.194	0.66	0.662	0.31	0.887	0	0.368	0.444	0.6	0.793	0.789
S_9	0	0.323	0.887	0.887	0.305	0.887	0.479	0.104	0.5	0.412	0.266	0.91
S_{10}	0.294	0.871	0.179	0.166	0.316	0.401	0.78	0.445	0.194	0.313	0.668	0.821
S_{11}	0.118	0.484	0.547	0.54	0.261	0.725	0.721	0.191	0.444	0.192	0.185	1
S_{12}	0.176	0.29	0.689	0.798	0.175	0.939	0.68	0.331	0.528	0.065	0	0.792
S_{13}	0.353	0.29	0.321	0.341	0.956	1	0.518	0.251	0	0.699	0.69	0.28
S_{14}	0.647	0.323	0.66	0.662	0.956	0.806	0.143	0.057	0.861	0.653	0.293	0
S_{15}	0.412	0.613	1	0.922	0.359	0.401	0.357	0.087	0.972	0.404	0.674	0.104
S_{16}	0.647	0.323	0.547	0.54	0.555	0.417	0.401	0.01	0.972	0.859	0.826	0.075
S_{17}	0.529	0.194	0	0	0.75	0.758	0.162	0.067	0.194	0.477	0.076	0.767
S_{18}	0.294	0.581	0.415	0.391	1	0.353	0.996	0.077	0.917	0.594	0.527	0.323
S_{19}	0.235	0.516	0.66	0.662	0.266	0.595	0.692	0.338	0.306	0.703	1	0.136
S_{20}	0.647	1	0.189	0.175	0.726	0.693	1	0.094	0.194	0.907	0.723	0.642
S_{21}	0.824	0	0.67	0.333	0.276	0.903	0.155	0.067	1	0.323	0.609	0.258
S_{22}	0.059	0.613	0.443	0.822	0.085	0.498	0.784	0.699	0.444	0.863	0.777	0.355
S_{23}	0.118	0.323	0.896	0.9	0.403	0.595	0.692	0.02	0.389	0.796	0.223	0.728
S_{24}	0.706	0.677	0.858	0.859	0.52	0.515	0.156	0.017	0.333	0.295	0.489	0.079
S_{25}	0.529	0.742	0.34	0.315	0.564	0.806	0.436	0.154	0.167	0.939	0.207	0.842
S_{26}	0.235	0.323	0.245	0.228	0.217	0.595	0.732	0.057	0.472	0.547	0.245	0.742
S_{27}	0.588	0.194	0.566	0	0.75	0.887	0.307	0.074	0.25	0.477	0.076	0.767

Table 6 (continued)

Sample sites	Chemical properties			Macronutrients				Micronutrients				
	pH	EC (ds/m)	OC %	*N* (Kg/ha)	*P* (Kg/ha)	K (Kg/ha)	Sulphur (ppm)	Zinc (ppm)	Boron (ppm)	Iron (ppm)	Manganese (ppm)	Copper (ppm)
S_{28}	0.235	0.71	0.057	0.053	0.202	0.773	0.781	0.023	0.361	0.547	0.696	0.659
S_{29}	0.353	0.613	0.708	0.709	0.334	0	0.385	0.495	0.444	1	0.228	0.81
S_{30}	0.412	0.645	1	1	0.359	0.401	0.43	0.087	1	0	0.674	0.09
Weightage	***0.15***	***0.05***	***0.20***	***0.15***	***0.14***	***0.14***	***0.05***	***0.03***	***0.02***	***0.03***	***0.02***	***0.02***
Weightage in %	***15***	***5***	***20***	***15***	***14***	***14***	***5***	***3***	***2***	***3***	***2***	***2***

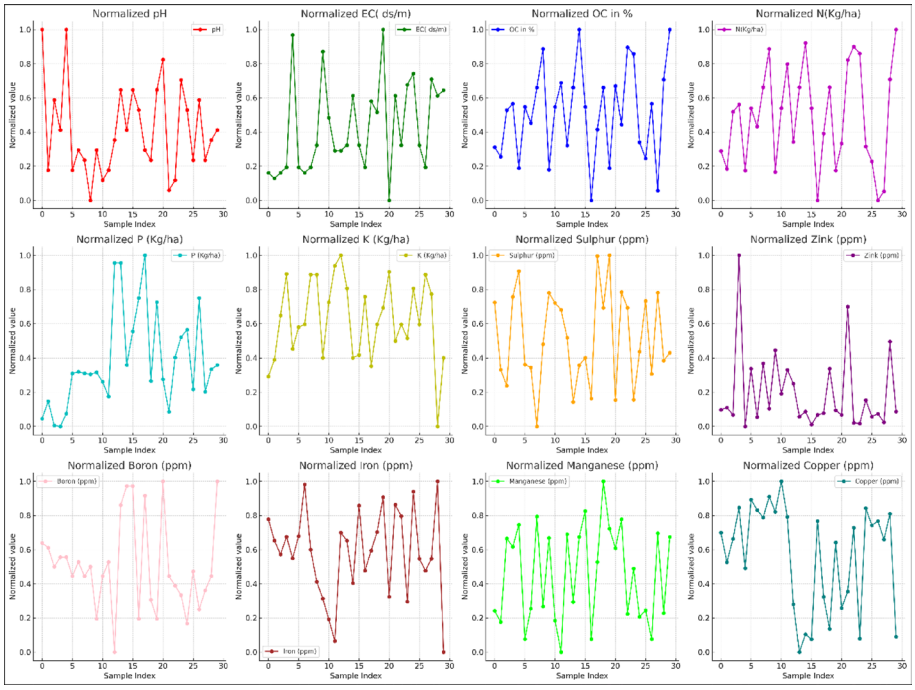


Fig. 7 Normalized data plot of different soil fertility parameters for SFI assessment of the study area

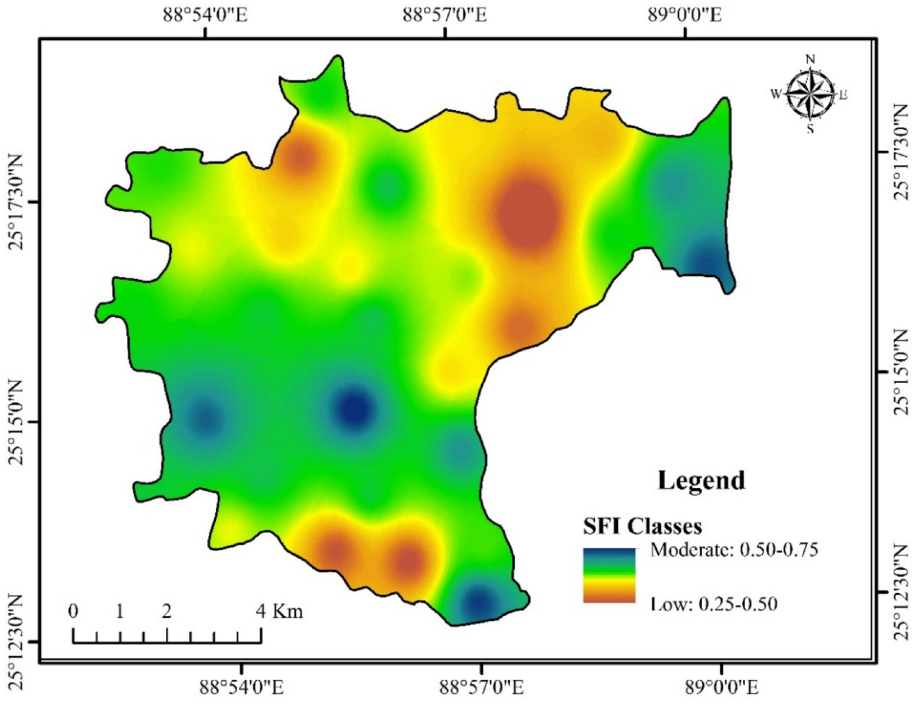


Fig. 8 Spatial variability map showing soil fertility index (SFI) distribution in the study area

are falling under a low fertility level as the soil is sandy or degraded soil that can hold fewer nutrients, low water retention capacity, Lack of biomass return (e.g., crop residue removal, deforestation) reduces humus content, Imbalanced use of NPK, and soil acidity hinders nutrient uptake [74, 129]. Besides that, Continuous cultivation without replenishment reduces the fertility level of those villages [34, 79]. Sarenbari, Chakgopal, Purba

Table 7 Soil fertility index (SFI) value and its associated fertility classes (Keshavarzi et al. 77; [105]

Soil fertility index value	SFI classes
0.00–0.25.00.25	Very low
0.25–0.50	Low
0.50–0.75	Moderate
0.75–0.90	High
0.90–1.00.90.00	Very high

Table 8 Site-specific SFI values and their associated fertility classes of the study area

Sample sites	Name of the villages	SFI value	SFI classes
S_1	Baigram	0.4	Low
S_2	Panjul	0.25	Low
S_3	Basanta	0.44	Low
S_4	Sarenbari	0.52	Moderate
S_5	Chakgopal	0.43	Low
S_6	Purba Kalikapur	0.43	Low
S_7	Chak Mohan	0.42	Low
S_8	Bhimpur	0.51	Moderate
S_9	Aptair	0.57	Moderate
S_{10}	Dabra	0.34	Low
S_{11}	Rupbati	0.45	Low
S_{12}	Lalpur	0.53	Moderate
S_{13}	Jantipur	0.53	Moderate
S_{14}	Matais	0.64	Moderate
S_{15}	Binsira	0.6	Moderate
S_{16}	Kaliha	0.52	Moderate
S_{17}	Gosaipur	0.35	Low
S_{18}	Gunjarpur	0.51	Moderate
S_{19}	Nafar	0.51	Moderate
S_{20}	Maniko	0.52	Moderate
S_{21}	Makhatair	0.53	Moderate
S_{22}	Purba Mustafapur	0.45	Low
S_{23}	Jamalpur	0.57	Moderate
S_{24}	Lakma	0.62	Moderate
S_{25}	Nawapara	0.5	Moderate
S_{26}	Fajullapur	0.33	Low
S_{27}	Purba Mathurapur	0.49	Low
S_{28}	Jamalpur	0.32	Low
S_{29}	Laskarpur	0.47	Low
S_{30}	Ujal	0.61	Moderate

Kalikapur, Chak Mohan, Bhimpur, Aptair, Dabra, Rupbati, Lalpur, Jantipur, Matais, Binsira, Kaliha, Gosaipur, Gunjarpur, Nafar, Maniko, Makhatair, Purba Mustafapur, Jamalpur, Lakma, Nawapara, Fajullapur, Purba Mathurapur, Jamalpur, Laskarpur, Ujal falling under medium fertility level because of these soils are often slightly to moderately acidic (pH < 6) due to leaching of bases and accumulation of iron and aluminum oxides, which makes nutrients like phosphorus and micronutrients less available [106, 120]. which was shown in Fig. 8.

The low fertility level of Hilli Block as the soil being rich in iron and aluminium oxides but lacking essential nutrients like nitrogen, phosphorus, and potassium, which are vital for plant growth [100, 172]. The high clay content in the soil leads to poor drainage, which can affect root development and nutrient uptake [180, 181]. The region has

low OC content, which means the soil lacks the humus needed to retain moisture and nutrients [37, 40]. Although the clay-rich soil of this region retains water, the structure often leads to waterlogging rather than the proper retention of moisture needed for crops [55, 57]. On the other hand, this region experiences a prolonged dry season, making it difficult for crops to thrive without irrigation as a supplement, and also the undulating topography and hard, crusted layers result in low water retention, affecting crop productivity and the fertility of the Hilli block [55, 141]. Therefore, in-situ moisture conservation techniques, such as mulching, contour bunding, and rainwater harvesting structures, can improve water availability and nutrient uptake [56]. Retaining crop residues or applying mulch also aids in maintaining soil temperature, preventing erosion, and contributing organic carbon [78, 130]. This means that the soil can support plant growth, but improvements are needed in specific parameters, such as adding organic amendments like compost or manure and using P and K-based fertilizers where deficiencies are evident to achieve optimal fertility [86, 115]. Addressing nutrient deficiencies by adjusting pH through lime or S applications based on the specific issue, and also maintaining pH balance, will be crucial in boosting soil fertility across the sites [3, 149, 172]. Regular monitoring and adopting sustainable soil management practices are recommended to ensure the long-term health and productivity of the soils [50, 79]. The study area requires soil rehabilitation techniques such as organic farming, using biofertilizers such as phosphate-solubilizing bacteria (PSB), Azotobacter, and Azospirillum, which play a vital role in improving nutrient uptake efficiency and promoting sustainable fertility enhancement [189]. Besides this, adopting crop rotation and improving irrigation systems to enhance fertility and support sustainable agriculture [61, 97].

4.2 Nutrient requirements mapping

4.2.1 Nitrogen (N) requirements

To achieve the maximum yield (5200 kg/ha), N requirements fluctuate across the sample sites, ranging from approximately 96 to 137 kg/ha for rice, which is depicted in Fig. 9.

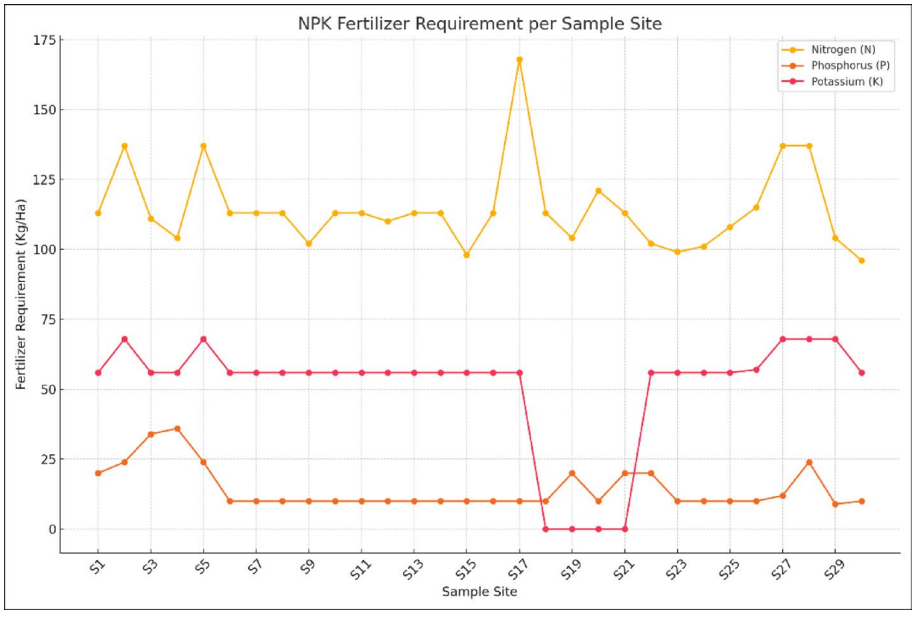


Fig. 9 Site-specific nutrient recommendation for the study area based on the model output results

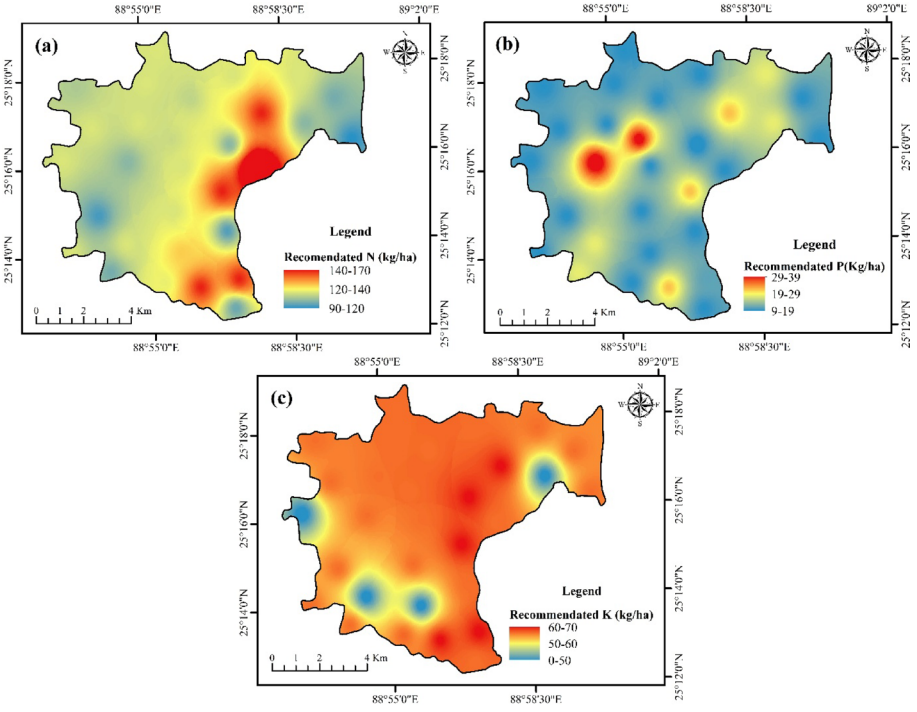


Fig. 10 Nutrient requirements mapping (**a**) N, (**b**) P, and (**c**) K for the study area

The highest nitrogen requirement is observed at sample sites S_{17} (168 kg/ha), while the lowest is at S_{30} (96 kg/ha). There seems to be a noticeable drop in nitrogen requirements, indicating the variability in nitrogen demand across different sites. The N requirement map (Fig. 10a) shows that Panjul, Chakgopal, Gosaipur, Fajullapur, and Purba Mathurapur villages are suggested to have high N requirements for rice cultivation.

4.2.2 Phosphorus (P) requirements

P requirements are generally lower than N, ranging between 9 and 36 kg/ha for rice cultivation. Sample site S_4 has the highest phosphorus requirement (36 kg/ha), while S_{29} shows the lowest (9 kg/ha) (Fig. 9). The highest P fertilizer rates are recommended in the central and western parts of the study area, consisting of the Basanta and Sarenbari areas which have been depicted in Fig. 10b.

4.2.3 Potassium (K) requirements

With a yield target of 5200 kg/ha, K requirements for rice-based cropping systems ranged from 56 to 68 kg/ha for rice cultivation (Fig. 9). Potassium requirements remain relatively stable across all sample sites. Gunjarpur, Nafar, Maniko, and Makhatair villages have no requirement of K-based fertilizer as this area already has a sufficient amount of K availability which has been shown in Fig. 10c.

5 Conclusion and recommendations

So far, the research question and objective are concerned, the present study proposes the following conclusions and recommendations:

- The study highlights spatial variation in soil fertility across the Barind tract, helping identify low-fertility and moderately fertile zones that require immediate attention and rehabilitation.
- The SNI and SFI revealed that 45.41% of the areas fall under low fertility and 54.49% under moderate fertility levels. These results highlight a general nutrient deficiency across the region. Soils in the region are acidic (average pH 5.4–5.6) with low organic carbon content (average 0.59–0.71%), indicating the need for soil amelioration through lime application (calcium carbonate or dolomitic lime at the rate of 3.71–7.41 t/ha) and Organic matter enrichment via the incorporation of vermicompost (2–4 t/ha), or farmyard manure (5–10 t/ha), Green manure crops like Sesbania or Dhaincha will naturally enhance nitrogen content in the soil through biological nitrogen fixation, reducing dependence on synthetic fertilizers of that region [92, 121]. Hence, cultivation of Dhaincha on a large scale has to be inspired as a policy matter to enhance the nitrogen level in the soil.
- Adopting the study's outcome for location-specific and scientific application of nutrients, ensuring efficient resource use is tried to assure here. FertiliCalc-based nutrient mapping for the study area indicated that, to achieve the expected yield per hectare, an optimal fertilizer application rate would be approximately 116 kg/ha of N, 22.5 kg/ha of P, and 62 kg/ha of K. As a practice of soil amelioration, Azolla is such an aquatic fern suitable for various environments and contains high protein that is rich in good Amino acids, vitamins, and minerals [6, 93]. As easily digestible by animals, it will enhance livestocking and dung production for bio manuring to develop community-based policy adaptation in the entire study area. This aligns with a maintenance of fertilization strategy and underscores the need for regular monitoring of soil health parameters in rice-based cropping systems in the Barind tract of West Bengal.
- The findings in gross indicate that to improve soil fertility in the poorly drained areas of the Hilli block, an integrated approach is essential, such as surface and subsurface drainage measures, like shallow ditches, land levelling, and deep ploughing, which can reduce the problem of waterlogging and will enhance infiltration which is a policy approach to be taken up by the local governance to recharge the groundwater too. Besides this, crop diversification with waterlogging-tolerant varieties (*Swarna-sub1*, *IR64-sub1*, *Jalashree*, *Sahbhagi Dhan*) should be introduced to support the root health, nutrient uptake, and maximum crop yields of the rice-based cropping system of the Barind Tract. Introduction of both endemic and high-yielding varieties will enhance agricultural efficiency. which is an approach to accomplish sustainable agriculture towards the management of food security as a policy matter.
- Farmers and policymakers should adopt in-situ moisture conservation techniques, such as mulching, crop residue retention to conserve moisture, and also promote rainwater harvesting to mitigate water stress and enhance nutrient use efficiency of the Barind tract.
- The soil fertility map and nutrient requirements mapping can serve as a decision-making tool for agronomic zoning, land capability classification, and soil conservation planning, and also, the outcome of this research will strengthen the effectiveness of schemes like the Soil Health Card, PM-Krishi Sinchayee Yojana, and Rashtriya Krishi Vikas Yojana (RKVY) through data-driven planning in the Barind

tract of Dakshin Dinajpur district. To facilitate the network, the Office of the Block Development Officer (CD block Hilli), District Agriculture department, Panchayet samitis, and Gram panchayets, etc., can be integrated for comprehensive GIS-based fertility Mapping.

- Future research on soil fertility should focus on temporal monitoring to assess long-term changes under varying land use and cropping systems. Integrating remote sensing and GIS technologies can enhance the spatial accuracy and prediction of fertility dynamics under the influence of climate change and anthropogenic stress.
- The study also has certain limitations, such as soil sampling not fully representing the heterogeneity of the Barind tract, and single-season data overlooking seasonal nutrient dynamics. The SFI construction involves subjective indicator selection and weighting, with limited validation against actual crop responses. Besides that, long-term nutrient balance and socio-economic feasibility were not fully considered for this study area.

Acknowledgements
The authors pay sincere thanks to the Pedological Laboratory of the Department of Geography and Applied Geography, University of North Bengal.

Author contributions
Sujoy Kumar Malo worked on Conceptualization, methodology, data collection, formal analysis, and original draft preparation and Snehasish Saha also did Supervision, validation, and critical revision of the manuscript for important intellectual content.

Funding
No funding has been provided by any funder.

Data availability
The corresponding author can provide the datasets created or analyzed during the current research upon reasonable request.

Declarations

Ethics approval and consent to participate
This work is primarily based on soil fertility and sustainable soil nutrient management, and it does not involve any information on human beings; thus, an ethical statement does not apply to the context of the manuscript.

Competing interests
The authors declare no competing interests.

Received: 25 January 2025 / Accepted: 15 October 2025
Published online: 19 October 2025

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